

Deepa Shree Chickballapur Venkatachalapathi

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EDUCATION

University of Wisconsin, Madison

Master of Science, Data Science (GPA: 3.62/4.0)

Sep 2024 - May 2026

Madison, WI

- **Achievements:** Merit-Based Scholarship
- **Coursework:** Statistical Methods, Inference and Modeling, Database Design, Optimization, Machine Learning, Deep Learning, HCI, Master's Research

PES University

Bachelor of Technology, Computer Science and Engineering (GPA: 9.24/10.0)

Aug 2019 - May 2023

Bengaluru, India

- **Achievements:** Prof. MRD Scholarship (Top 5% students)
- **Coursework:** Data Structures and Algorithms, Operating Systems, Linear Algebra, Data Analytics, Cloud Computing, Big Data Systems, Distributed Systems

EXPERIENCE

University of Wisconsin-Madison

Student Data Analyst

2024 - Present

Madison, WI

- Reduced Operational Costs by 5% with dashboard development for data collected across various departments on the campus using Tableau, SQL, Microsoft Excel and SmartSheet (for project planning).
- Improved dashboards accessibility by developing automatic alternative text generation for tagging using Python and Tableau, enhancing user experience for visually impaired users.

Amazon Web Services

Software Engineer Intern

May 2025 - Aug 2025

Seattle, WA

- Developed ETL pipelines and data models for terabyte-scale AWS Host vulnerability data using SQL, PySpark & AWS services (S3, Glue, Redshift, Athena).
- Automated infrastructure & workflows with CloudFormation, AWS secure-cdk & Typescript for scalable data processing.
- Built near real-time dashboards with CloudWatch, SNS, and Lambda, improving host security visibility by 10%.

University of Wisconsin-Madison

Graduate Research Assistant

Feb 2025 - May 2025

Madison, USA

- Fine-tuned PyTorch vision models on HTC clusters to classify cranberry growth stages for accurate nutrient supply.
- Used roboflow to automate annotation of cranberries to support object detection to improve prediction accuracy by 7%.

Hewlett Packard Enterprise

Cloud Developer - I

Aug 2023 - Aug 2024

Bengaluru, India

- Optimized metadata retrieval with natural language support for CMF (MLOps library) with RAG Pipelines using Python, Hugging Face and Pinecone.
- Created a voice-based UI navigation feature by integrating STT and TTS with a language model core using React.js and Python, improving user interaction.
- Increased impression rate by 23% with UI and dashboard development using React.js, Grafana and Humio.
- Reduced database fetch time by 80% with caching optimization using Python, PostgreSQL, and Redis.
- Automated error logging pipelines using GitHub Actions and Terraform, reducing latency by 90%.

Hewlett Packard Enterprise

Research and Development Intern

Jan 2023 - Jul 2023

Bengaluru, India

- Automated regression testing using JUnit & Jenkins, reducing manual effort and improving deployment speed by 30%.
- Implemented a streaming error log parser in Python using NLP techniques to process gigabyte sized log files on SFTP servers, reducing triage time by 80%.

Adobe Systems

Technical Consultant Intern

May 2022 - Aug 2022

Bengaluru, India

- Developed an incremental ML pipeline for predicting purchase propensity using Python, achieving an F-measure over 0.9.
- Implemented interactive dashboards using MySQL and Power BI, increasing reporting efficiency by 15%.

PROJECTS

Reward Hacking Detection in Language Models | Interpretability, RLHF, Pytorch, CUDA | Advisor: Prof Sharon Li

- Embedding-based retrieval approach to identify deceptive or reward-hacked outputs efficiently and accurately.

Automating Analysis for Data Cubes | Databases, SQL, C++ | Advisor: Prof. Goetz Graefe

- Designed SQL queries and efficient, scalable approach for indexing & incrementally maintaining the cube and its indexes to ensure cost reduction for maintaining materialized views.

Catalog Indexing Engine | LangChain, Kafka, ElasticSearch, PostgreSQL, sklearn, Docker, Flask

- Designed a containerized agent system for catalog search and retrieval, integrating APIs with real-time streaming, re-indexing, and incremental learning to ensure product relevancy.

TECHNICAL SKILLS

- **Programming Languages & Scripting:** Python, C++, JavaScript, Java, C, R, Linux CLI
- **Data Management & Analysis:** SQL, MongoDB, Tableau, PowerBI, PySpark, Hadoop, Kafka, dbt, Airflow
- **Machine Learning:** Tensorflow, Keras, PyTorch, HuggingFace, LangChain, Pinecone, NLTK, scikit-learn, pandas, numpy
- **Cloud, Development & Monitoring Tools:** Docker, Kubernetes, AWS, React.js, Git, Github Actions, Terraform, Grafana